

Supplementary material for: Auditing crop host supervision in Swin models for condition recognition across multiple crops: evidence from 90 controlled runs

Muhammad Omair Khan^a, Muhammad Ahmad^a, MD Ariful Islam
Mozumder^a, Hee-Cheol Kim^b

^a*Institute of Digital Anti-Aging Healthcare, Inje University, Gimhae 50834, Republic of Korea.*

^b*Department of Computer Science and Engineering, Inje University, Gimhae 50834, Republic of Korea.*

S1. Experimental identity and computing environment

The final study matrix contained 90 completed Swin-S fine-tuning runs: 10 paired flat and dual runs per dataset and five matched runs per dataset for each of the MAE and $KL(q||p^h)$ consistency variants. The primary seeds were 2026–2035; the penalty seeds were 2026–2030. Every retained run completed 30 epochs and used validation macro-F1 for checkpoint selection.

The pretrained backbone was identified by the following records:

`swin_small_patch4_window7_224.ms_in22k_ft_in1k`

Hugging Face revision:

`221ee757828e4772df860bd42ef476de78dd5fc6`

Cached `model.safetensors` SHA-256:

`e17adf843764761f138cb8cab081d9a37824db4a948c0082c7af7f0092d19581`

The supplied protocol records Python 3.11.14, PyTorch 2.9.0+cu129, and an NVIDIA GeForce RTX 5070. Training used CUDA automatic mixed precision with gradient scaling, while selected model weights and final primary logits were stored in FP32.

Table S1: Locked split-manifest identities recorded by the experiment protocol.

Dataset	SHA-256
NPD-Rice-COCO-43	<code>de4fd19299c735a4ab939d67288f01757822a671cfb8d6c41b95f71d5fdd34ae</code>
MultiCrop-88	<code>b881f3e7feff8006f03a4d1e0066abf88cddda79dbee9295be3157cb11842231</code>
Agri-Foundation-145k	<code>ae87397218c8124721117e401291f10a3426ec0f94791b155a24fe3245c10732</code>

A permanent reproducibility repository is available in the GitHub repository. The repository preserves the protocol, manifest and taxonomy identities, run-completion records, derived seed-level metrics, statistical outputs, and figure-generation scripts.

S2. Dataset construction and integrity details

NPD-Rice-COCO-43.. The 93,388 paths represented 93,362 unique SHA-256 hashes and 51,758 source families. The local composite comprised 87,425 paths from the New Plant Diseases collection, 3,355 Rice Leafs images, and 2,608 selected COCO images used as a supervised background class. Twenty-six duplicate-hash clusters were retained, but every exact match and every identified augmentation family was assigned wholly to one split. The final split contained 74,700 training, 9,355 validation, and 9,333 test paths; the test set contained 5,171 source families. Every class occurred in every split. Assignment-group, source-family, and exact SHA-256 intersections were zero for every pair of partitions.

MultiCrop-88.. The input inventory contained 79,086 paths. We removed 1,254 redundant paths from 1,232 same-label exact-hash clusters and excluded all 62 paths from 29 cross-label conflict clusters. The 77,770 retained hashes were unique. The resulting split contained 62,214 training, 7,787 validation, and 7,769 test images, with all 88 classes represented in every partition. The local class vocabulary strongly matches the published 91-class composite of Ahamed et al. after exclusion of three mulberry classes, but the original assembly manifest and licenses were not supplied. The collection is therefore described as a local derivative. Its manifest `family_id` is an exact-hash assignment unit, not a separately verified acquisition family.

Agri-Foundation-145k.. The public Zenodo record reports 144,751 images in 215 classes. The locked local derivative contained 144,752 paths: a development pool later divided into 92,571 training and 23,140 validation images, plus a separate 29,041-image test set. The one-path difference is a synthetic 10×10 file named `dummy.jpg` in class `cabbage_cabbage_black_rot_bing_-_copy`. It was the only development support for that joint class. The file is not an observed plant image and must not be interpreted as biological evidence. The exact-hash audit found no development-to-test match. All 215 classes occurred in training and test; 212 occurred in validation.

The locked results retain the placeholder-supported class so that every configuration is evaluated on the experiment that was actually run. Because all models use the same derivative, seed-paired architecture comparisons remain case matched. Nevertheless, the class’s absolute F1 and its contribution to 215-class macro-F1 are not representative of ordinary learned performance. The source archive contains a script named `ds2_placeholder_sensitivity.py` that recomputes accuracy and macro-F1 after excluding this class. The numerical outputs of that script were not present in the supplied archive and are therefore not reported or inferred here. They should be generated, checked, and deposited before submission.

Exact byte hashing cannot detect relationships created only by resizing, recompression, cropping, color adjustment, or other transformations. No perceptual-hash or embedding-similarity overlap audit was available for MultiCrop-88 or Agri-Foundation-145k. The supported claim is therefore exact identity control, plus source-family control only for NPD-Rice-COCO-43.

S3. Pairing, equivalence decisions, and event-scale context

Flat and no-penalty dual models used identical seed sets (2026–2035) within each dataset. The primary equivalence margin was ± 0.25 percentage points. A dataset and metric result was considered practically equivalent only when its complete 90% TOST interval lay inside the margin and the Holm-adjusted decision passed at $\alpha = 0.05$ across the nine primary decisions. Five cells met both requirements: accuracy, macro-F1, and CHER on NPD-Rice-COCO-43, plus accuracy and CHER on MultiCrop-88.

The complete nine-decision Holm correction was repeated post hoc at ± 0.10 and ± 0.50 points. At the stricter margin, NPD-Rice-COCO-43 accuracy and CHER and MultiCrop-88 CHER remained equivalent. At ± 0.50 points, Agri-Foundation-145k accuracy and CHER also met equivalence, while macro-F1 remained unresolved on MultiCrop-88 and Agri-Foundation-145k. These analyses show which conclusions are robust to the choice of negligible-effect threshold.

Mean flat/dual cross-host event counts were 3.5/2.2 of 9,333 test images, 26.1/23.8 of 7,769 images, and 964.5/911.4 of 29,041 images for the three datasets. A 0.10-point CHER margin corresponds to 9.3, 7.8, and 29.0 events; a 0.25-point margin corresponds to 23.3, 19.4, and 72.6 events; and a 0.50-point margin corresponds to 46.7, 38.8, and 145.2 events. All intervals

characterize between-seed variation on fixed test sets, not sampling variation over new populations.

S4. Family-balanced NPD-Rice-COCO-43 sensitivity analysis

Each NPD-Rice-COCO-43 test image was assigned weight equal to the inverse of its source-family size, giving every one of the 5,171 test families total weight one. Flat/dual family-balanced means were 99.033/99.020% accuracy, 98.522/98.491% macro-F1, and 0.0216/0.0107% CHER. Paired dual-minus-flat effects were -0.0129 points (95% CI $[-0.0829, 0.0571]$), -0.0307 points ($[-0.1352, 0.0739]$), and -0.0110 points ($[-0.0241, 0.0022]$), respectively. The near-null result therefore persisted when each original family, rather than each augmented path, had equal total influence.

S5. Consistency-penalty implementation and validation

The secondary block contained five matched seeds for the no-penalty dual model, MAE, and $\text{KL}(q||p^h)$ on each dataset. The dual objective used condition cross-entropy plus host cross-entropy with $\alpha = 1$. Both consistency terms used $\lambda = 1$. MAE was averaged over samples and host categories. KL was implemented as $D_{\text{KL}}(q||p^h)$, summed over host categories and averaged over samples, with $\epsilon = \text{torch.finfo}(p^h.\text{dtype}).\text{eps}$ in both logarithms. The condition-implied host distribution q was not detached, so the consistency gradient passed through both the condition and host branches.

The available validators checked run completion, 30-row training histories, selected checkpoints, split identities, and saved test predictions. Stored best validation macro-F1 values were checked against the corresponding histories, and accuracy, macro-F1, and CHER were reconstructed from saved labels and standard predictions where available. The fixed unit coefficients were not selected from locked-test performance.

S6. Complete head-disagreement and TCI results

HDR compares the host implied by the condition argmax with the direct host-head argmax. TCI was implemented by masking condition classes to the host predicted by the direct host head. The operation reduced post-TCI HDR to zero in every one of the 60 dual-head runs by construction. Table S2 reports the complete configuration-level means. Flat runs are absent because they have no independent host prediction.

Table S2: Complete head-disagreement and taxonomic consistency inference (TCI) results; TCI was implemented as predicted-host masking. Host accuracy and pre-TCI head disagreement rate (HDR) are percentages reported as mean $\pm$ standard deviation over matched final-protocol runs. Deltas are TCI minus standard inference in percentage points; **CHER** denotes cross-host error rate.

Dataset	Configuration	Seeds	Host accuracy	Pre-TCI HDR	Δ accuracy	Δ macro-F1	Δ CHER
NPD-Rice-COCO-43	Dual, no penalty	10	99.977 $\pm$ 0.003	0.004 $\pm$ 0.006	0.001 $\pm$ 0.003	0.001 $\pm$ 0.003	-0.001 $\pm$ 0.003
NPD-Rice-COCO-43	Dual + MAE	5	99.976 $\pm$ 0.005	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
NPD-Rice-COCO-43	Dual + KL($q p^h$)	5	99.976 $\pm$ 0.005	0.002 $\pm$ 0.005	0.002 $\pm$ 0.005	0.002 $\pm$ 0.004	-0.002 $\pm$ 0.005
MultiCrop-88	Dual, no penalty	10	99.716 $\pm$ 0.025	0.102 $\pm$ 0.026	0.013 $\pm$ 0.033	0.033 $\pm$ 0.116	-0.022 $\pm$ 0.037
MultiCrop-88	Dual + MAE	5	99.709 $\pm$ 0.035	0.072 $\pm$ 0.027	-0.005 $\pm$ 0.020	-0.009 $\pm$ 0.070	0.008 $\pm$ 0.020
MultiCrop-88	Dual + KL($q p^h$)	5	99.745 $\pm$ 0.028	0.067 $\pm$ 0.011	0.018 $\pm$ 0.030	0.100 $\pm$ 0.147	-0.021 $\pm$ 0.023
Agri-Foundation-145k	Dual, no penalty	10	96.999 $\pm$ 0.047	1.144 $\pm$ 0.052	0.016 $\pm$ 0.036	0.169 $\pm$ 0.376	-0.137 $\pm$ 0.050
Agri-Foundation-145k	Dual + MAE	5	96.892 $\pm$ 0.113	1.148 $\pm$ 0.044	-0.002 $\pm$ 0.039	0.079 $\pm$ 0.207	-0.100 $\pm$ 0.044
Agri-Foundation-145k	Dual + KL($q p^h$)	5	97.018 $\pm$ 0.035	0.861 $\pm$ 0.047	-0.014 $\pm$ 0.046	-0.003 $\pm$ 0.266	-0.052 $\pm$ 0.031

For the 10 no-penalty Agri-Foundation-145k dual runs, TCI changed a mean of 332.3 predictions per seed. Among those disagreements, the host head alone matched the labeled host in 38.7%, the condition-implied host alone matched it in 26.8%, and both were wrong in 34.5%. TCI gained 79.1 and lost 74.4 correct top-1 decisions per seed, leaving a mean net gain of 4.7 cases.

S7. Long-tail, host-level, oracle, and calibration evidence

The 215 Agri-Foundation-145k classes were grouped by training support into 1–25, 26–100, 101–500, and more than 500 images. Mean dual-minus-flat within-stratum macro-F1 changes were 0.871, 0.600, 0.854, and 0.035 points, respectively. Only the 101–500 stratum had a paired 95% interval wholly above zero. These comparisons are exploratory and were not multiplicity adjusted. The largest positive and negative class changes generally occurred in classes with single-digit test support.

Labeled-host oracle masking was applied to fixed condition logits without retraining. It increased mean accuracy by 2.294 points for the flat model and 2.184 points for the dual model; macro-F1 increased by 10.601 and 10.232 points, respectively. The oracle is non-deployable and sets CHER to zero by construction. It estimates the error removable by perfect host information rather than the performance of a practical host sensor.

For every primary flat and dual checkpoint, one temperature was fitted only on validation logits and then applied to locked-test logits. Temperature scaling reduced ECE, NLL, and summed multiclass Brier score in all six dataset

and configuration cells. AURC changed little, as expected for a monotonic single-temperature transformation that largely preserves confidence ranking.

S8. Repository contents required before submission

The permanent repository contains, at minimum:

- exact split manifests, ordered class lists, condition-to-host mappings, and their SHA-256 identities;
- the frozen experiment protocol, environment record, and run-completion metadata for all 90 runs;
- seed-level primary metrics and paired differences, TOST/Holm outputs, margin-sensitivity results, and CHER event-count calculations;
- family-balanced, penalty, HDR/TCI, disagreement-decomposition, long-tail, host-level, labeled-host oracle, calibration, and risk-coverage outputs;
- the executed Agri-Foundation-145k placeholder-class sensitivity results, including the audit record for the 10×10 file;
- scripts used to train models, validate runs, regenerate tables and figures, and recompute checksums; and
- a concise README that maps every manuscript table and figure to its generating files and command.

The supplied archive contains source code and manuscript assets but does not contain every machine-readable result file named in earlier draft notes. Accordingly, this supplementary document describes only evidence supported by the supplied text, tables, figures, configuration files, and scripts; it does not claim that missing repository artifacts are already attached.